Marketing Campaign Response — Summary

Predicting who responds to a campaign, and turning it into a targeting and budget plan.

▶ Open the live app   Code & notebook on GitHub

2,237	**14.9%**	**XGBoost**	**0.90**	**~3×**
CUSTOMERS	BASELINE RESPONSE	BEST MODEL	ROC-AUC	WARM-LIST LIFT

Bottom line. A retailer's last campaign reached everyone, but only ~15% responded, so most of the spend was wasted. The model scores each customer's likelihood to respond (ROC-AUC ≈ 0.90). Aimed at the right people, the same budget earns far more.

Approach

Cleaned 2,237 customer records (filled 24 missing incomes, removed a few typo rows, tidied messy labels), then built 10 behaviour features: spending, recency, engagement, past-campaign history, income tiers. Trained and compared three models inside leak-free scikit-learn pipelines with 5-fold cross-validation. Because only ~15% respond, models are judged on ranking quality (ROC-AUC) and recall, not plain accuracy, and each is told to weight the rare "yes" cases.

Key insights

- **Past responders are gold:** people who accepted a past campaign respond at **41%** vs **8%** (~3×).
- **Higher income, higher response: 28%** vs **10%**; high income + a past acceptance reaches **47%.**
- **Recent buyers are warm:** responders last bought ~35 days ago vs ~52 for everyone else.
- **Households without kids respond more: 27%** vs **10%** with children.
- **Catalog buyers over-respond** (4.2 vs 2.4 purchases); website *visits* don't predict response at all.

Model performance

Model	Acc	Prec	Rec	F1	AUC
XGBoost ★	**0.89**	**0.63**	**0.61**	**0.62**	**0.90**
Random Forest	0.89	0.72	0.42	0.53	0.89
Logistic Reg.	0.81	0.42	0.73	0.53	0.88

XGBoost wins on ranking (ROC-AUC) and F1. Since a missed responder (~$11) costs more than a wasted contact (~$3), we lean toward recall and set the contact cutoff where expected profit is highest, not at a textbook 0.5.

Business recommendations

- **Contact first:** rank everyone by predicted probability; start with the ~460 prior responders (~3× the base rate, free to find).
- **Budget:** concentrate on the top 2–3 probability deciles; stop funding the unresponsive bottom half.
- **Personalise:** premium offers for high-income responders, recency-triggered sends for warm buyers, sustained catalog investment.
- **Prove it:** track profit per contact against a random-targeting control so the lift is provable in dollars.

Live app: https://marketing-campaign-response.streamlit.app · **Code:** github.com/RABI9000/marketing-campaign-response · **Full analysis:** Jupyter notebook

Built with Python, pandas, scikit-learn, XGBoost, and Streamlit.